Phase 4 — Web tier, launch template, ALB, target group, and Auto Scaling

The web tier is based on a launch template and an Auto Scaling Group named cloudshield-web-asg. The Application Load Balancer is internet-facing and spans both public subnets. The target group cloudshield-web-tg registers the web instances and uses its configured health check to determine target health.

Why this phase matters

The ALB gives users a single entry point and distributes requests between web instances. Running web capacity across two AZs reduces dependence on one AZ. The ASG can maintain the configured capacity and adjust the number of instances according to the scaling policy.

What was configured

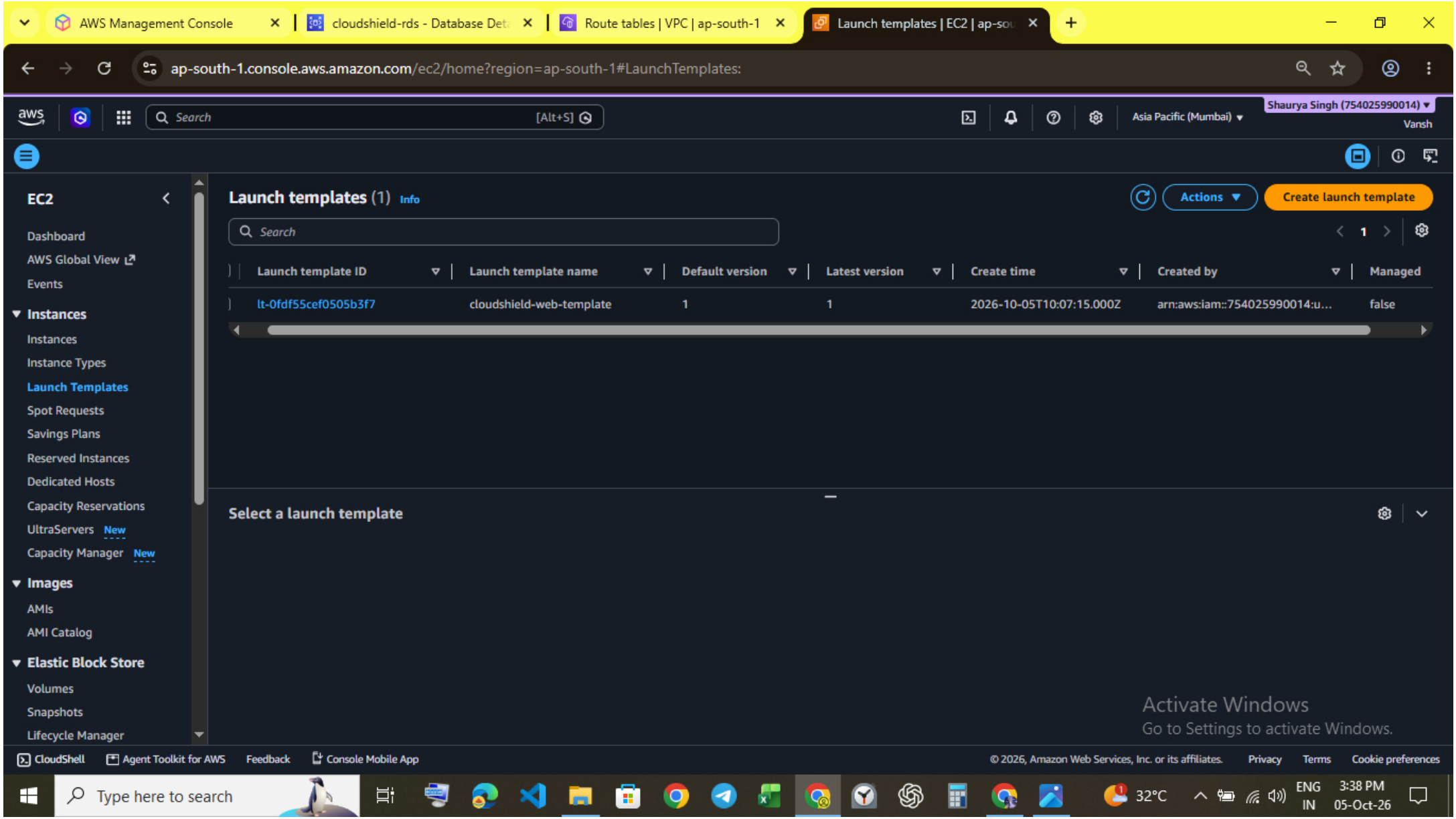
• The final diagram shows two web EC2 instances, one in each public AZ.

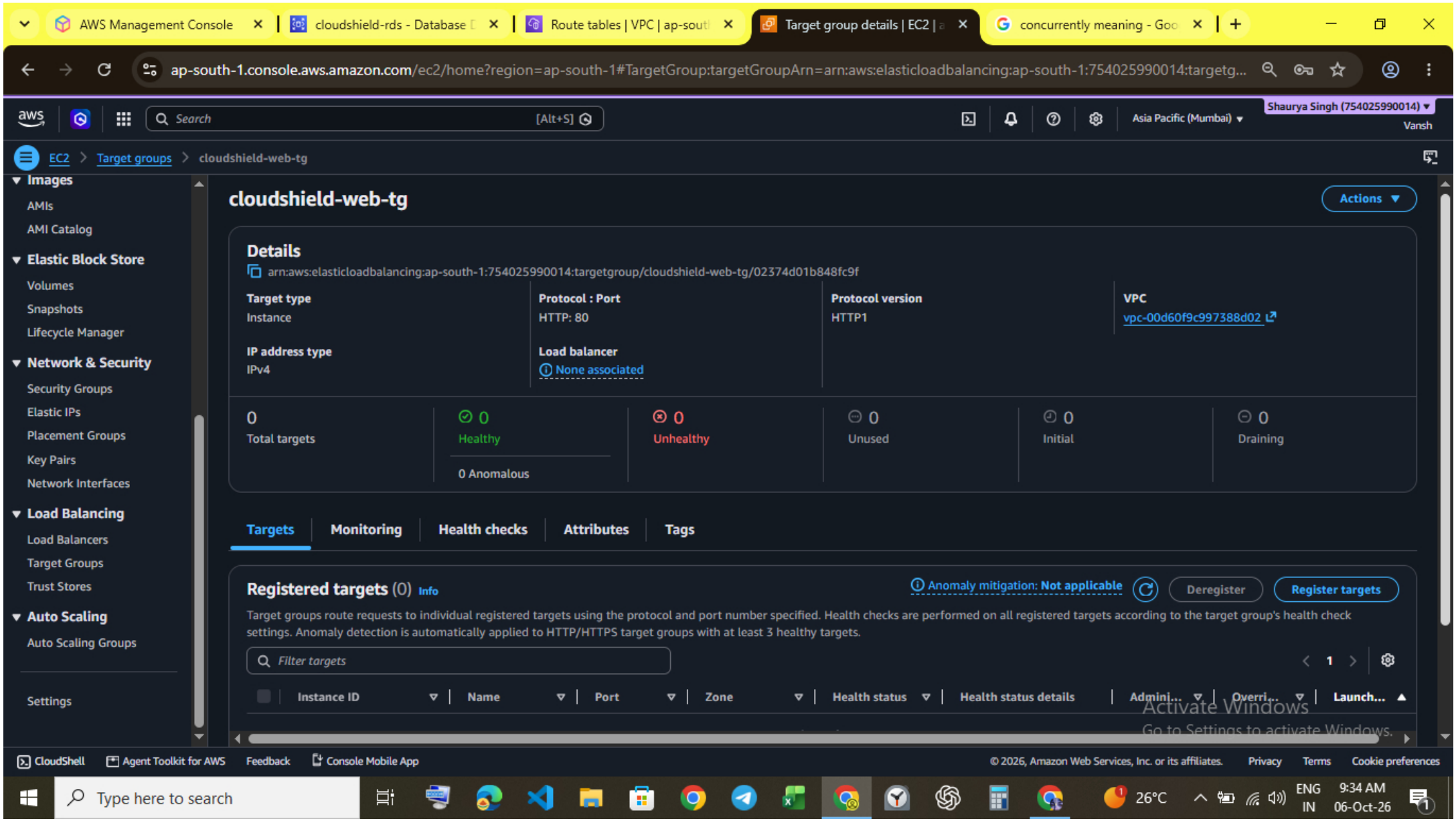
• The diagram records ASG minimum 2, maximum 3, and CPU target tracking at 60%.

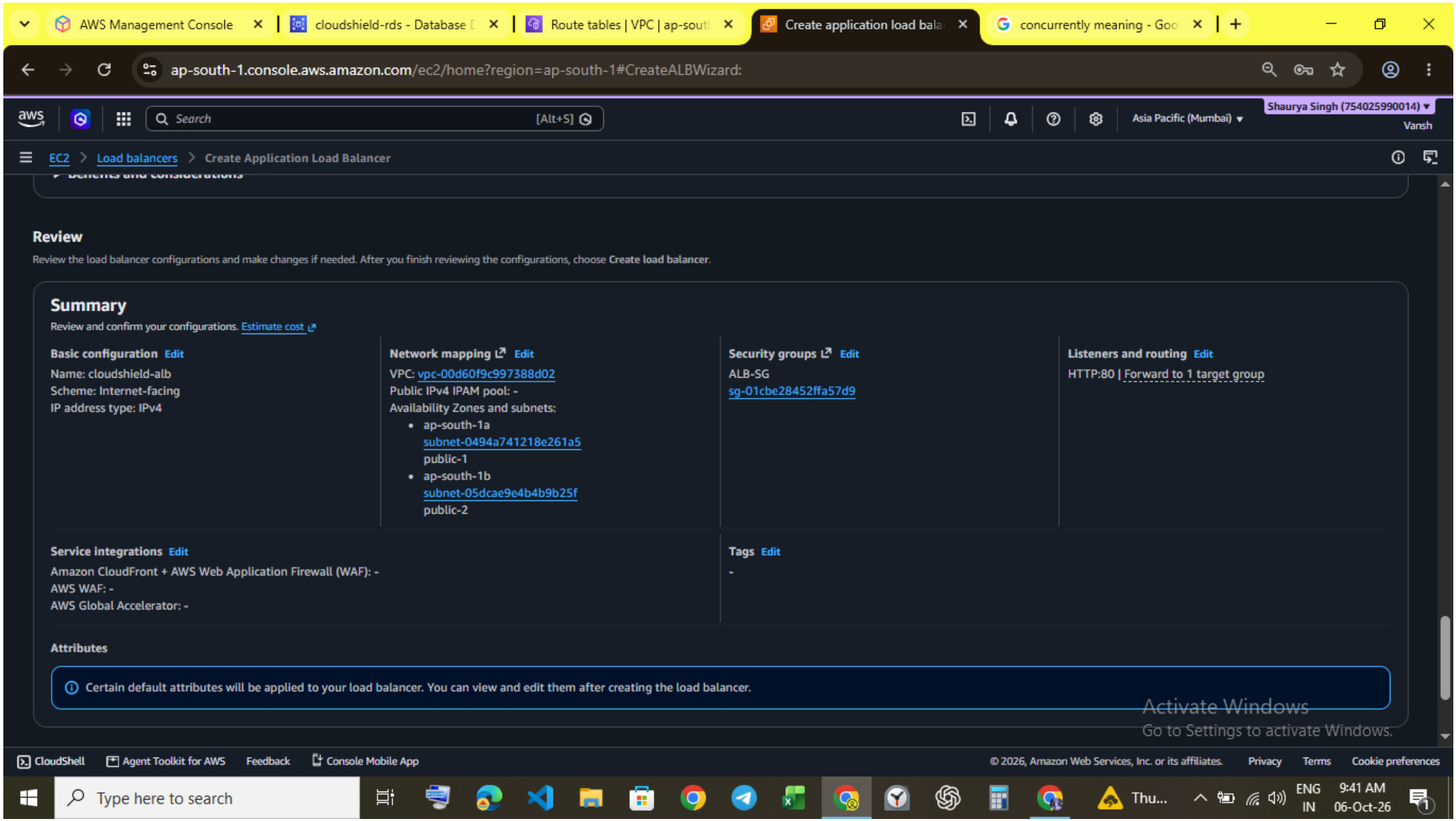
• The ALB is labelled HTTP 80 and forwards traffic to cloudshield-web-tg.

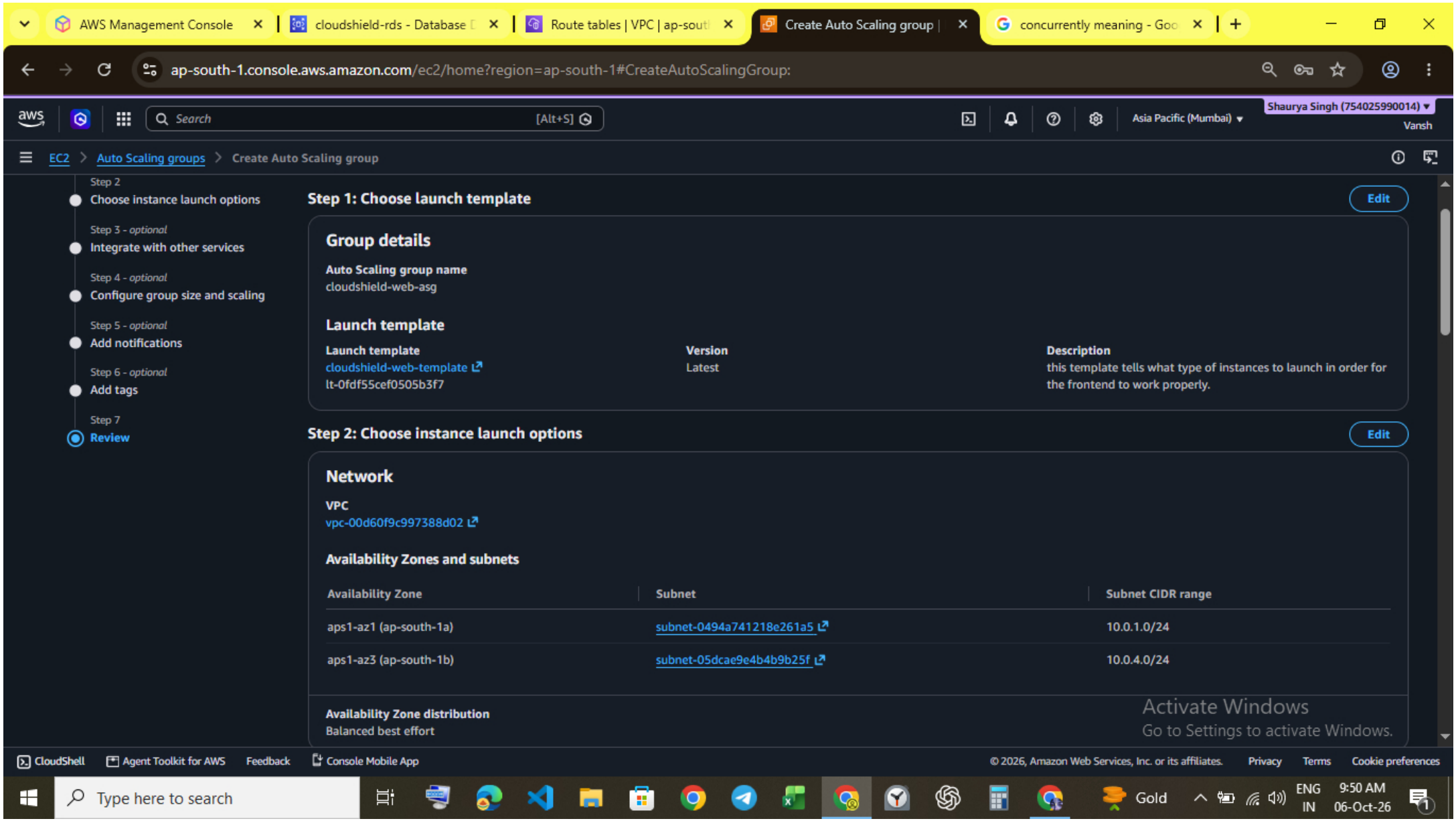
• The backend tier is shown as a separate tier; the diagram labels web-to-backend traffic on port 3000.

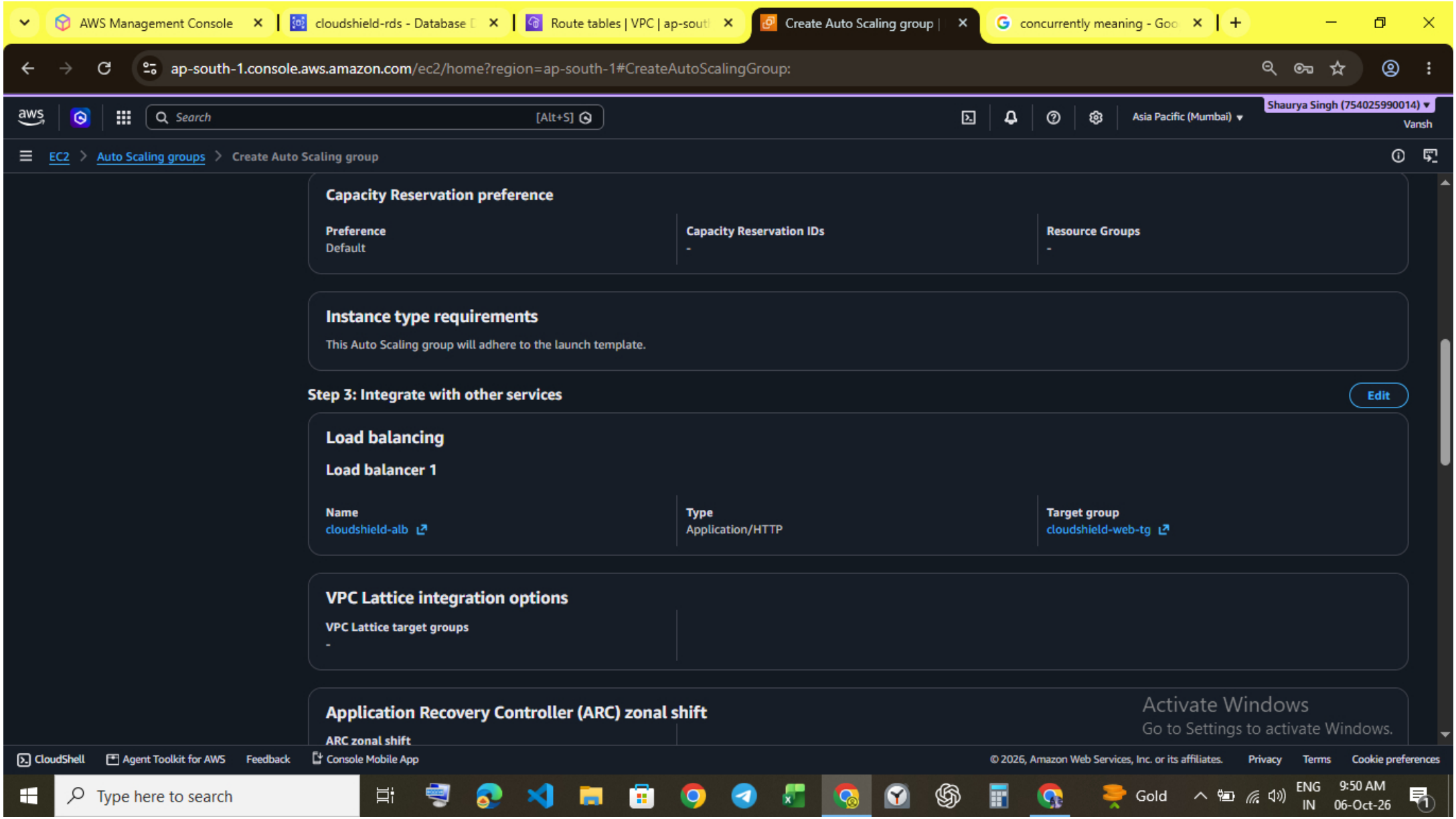
> The screenshots immediately following this phase are original pages 61–76, preserving the launch template, load balancing, target group, and scaling setup sequence.

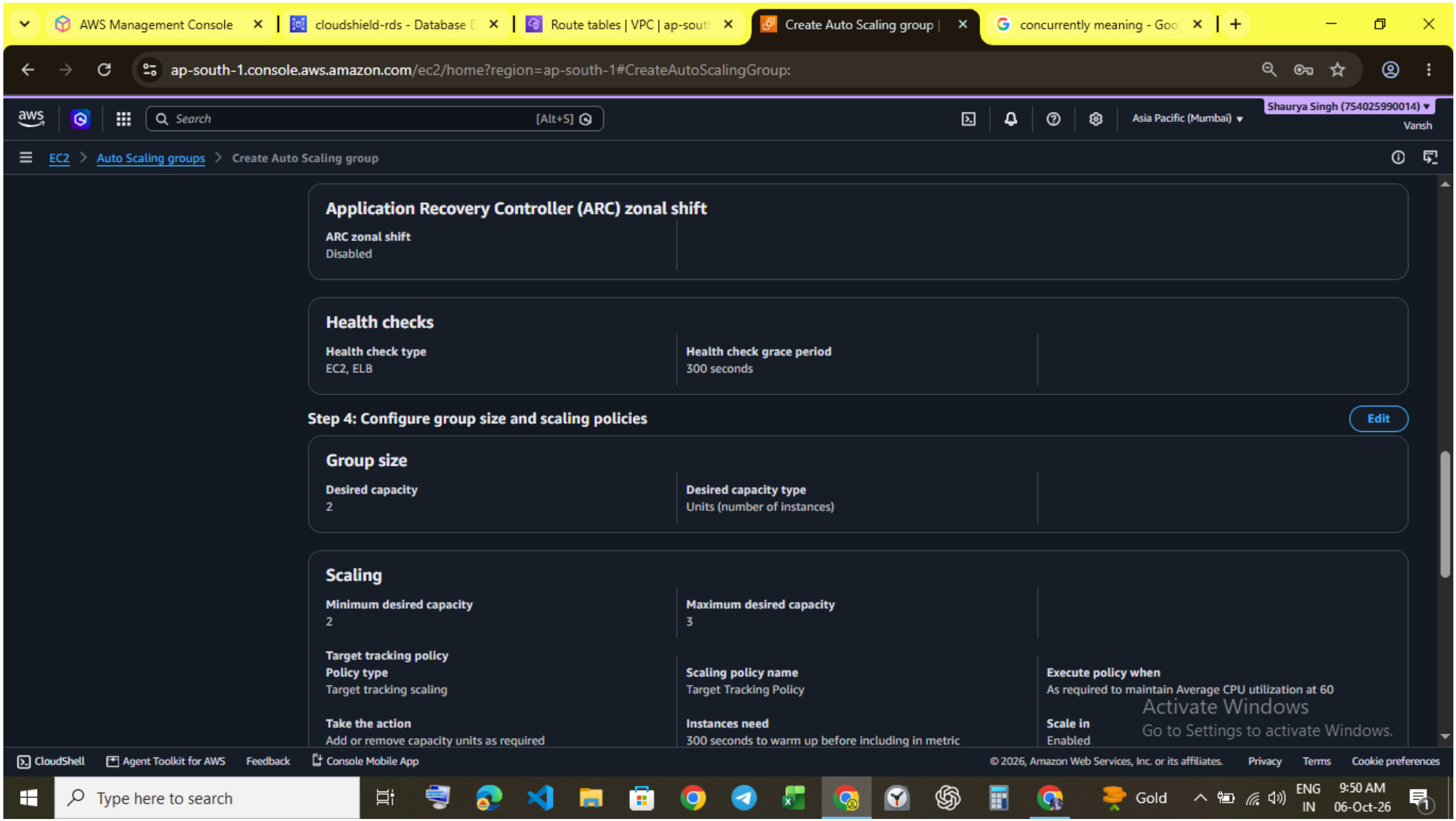

AWS Management Console
cloudshield-rds - Database Det
Route tables | VPC | ap-south-1
Launch templates | EC2 | ap-sou
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchTemplates:
Search
[Alt+S]
Asia Pacific (Mumbai)
Shaurya Singh (754025990014)
Vansh
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Launch templates (1) Info
Actions
Create launch template
Search
1
Launch template ID
Launch template name
Default version
Latest version
Create time
Created by
Managed
lt-0fdf55cef0505b3f7
cloudshield-web-template
1
1
2026-10-05T10:07:15.000Z
arn:aws:iam::754025990014:u...
false
Select a launch template
Activate Windows
Go to Settings to activate Windows.
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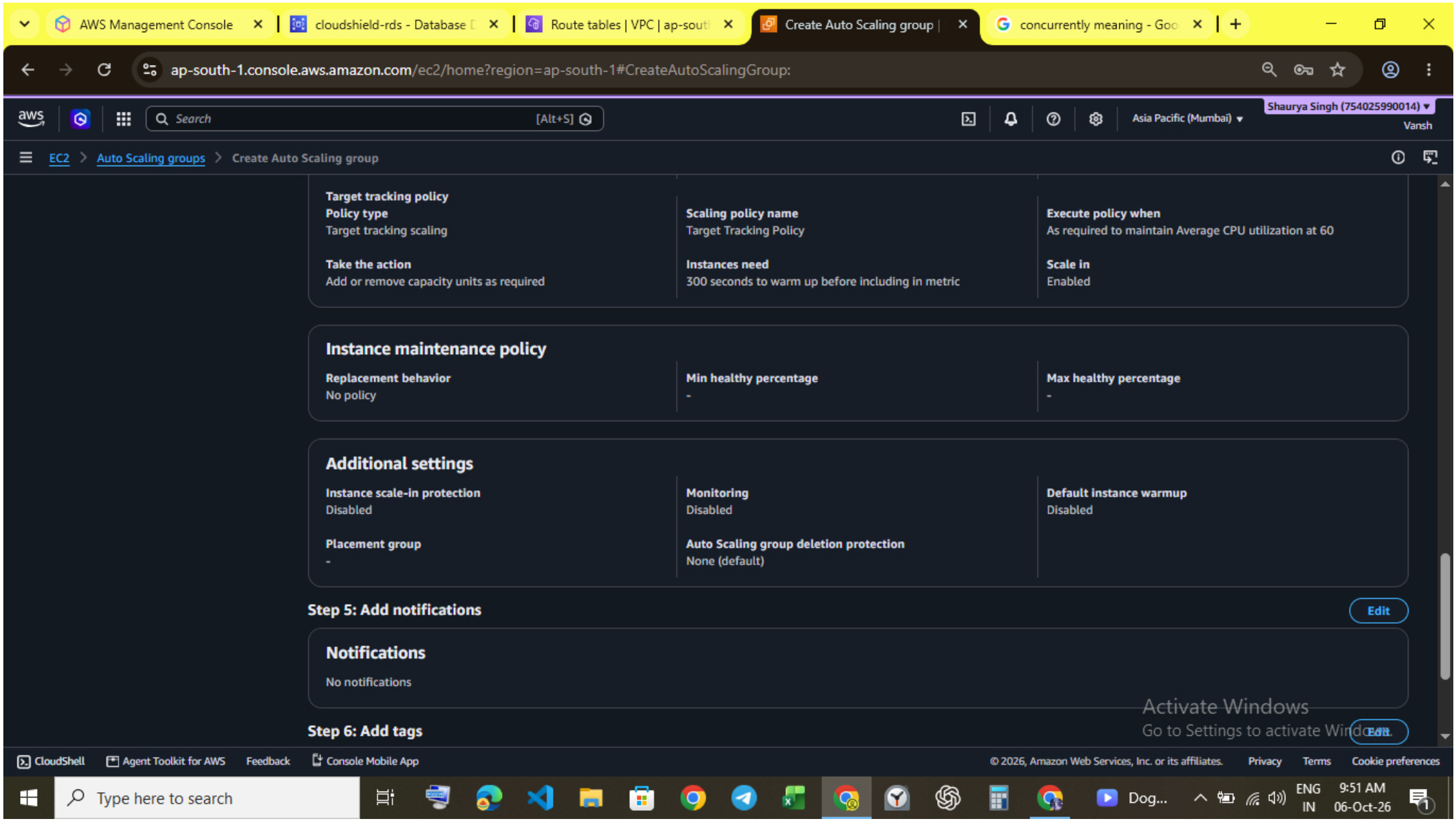

AWS Management Console
cloudshield-rds - Database
Route tables | VPC | ap-sout
Target group details | EC2 |
concurrently meaning - Goo
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#TargetGroup:targetGroupArn=arn:aws:elasticloadbalancing:ap-south-1:754025990014:targetg...
Search [Alt+S]
Asia Pacific (Mumbai)
Shaurya Singh (754025990014)
Vansh
EC2 > Target groups > cloudshield-web-tg
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
cloudshield-web-tg
Actions
Details
arn:aws:elasticloadbalancing:ap-south-1:754025990014:targetgroup/cloudshield-web-tg/02374d01b848fc9f
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-00d60f9c997388d02
IP address type
IPv4
Load balancer
None associated
0
Total targets
0
Healthy
0 Anomalous
0
Unhealthy
0
Unused
0
Initial
0
Draining
Targets
Monitoring
Health checks
Attributes
Tags
Registered targets (0) Info
Anomaly mitigation: Not applicable
Deregister
Register targets
Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.
Filter targets
Instance ID
Name
Port
Zone
Health status
Health status details
Admini...
Overri...
Launch...
Activate Windows
Go to Settings to activate Windows.
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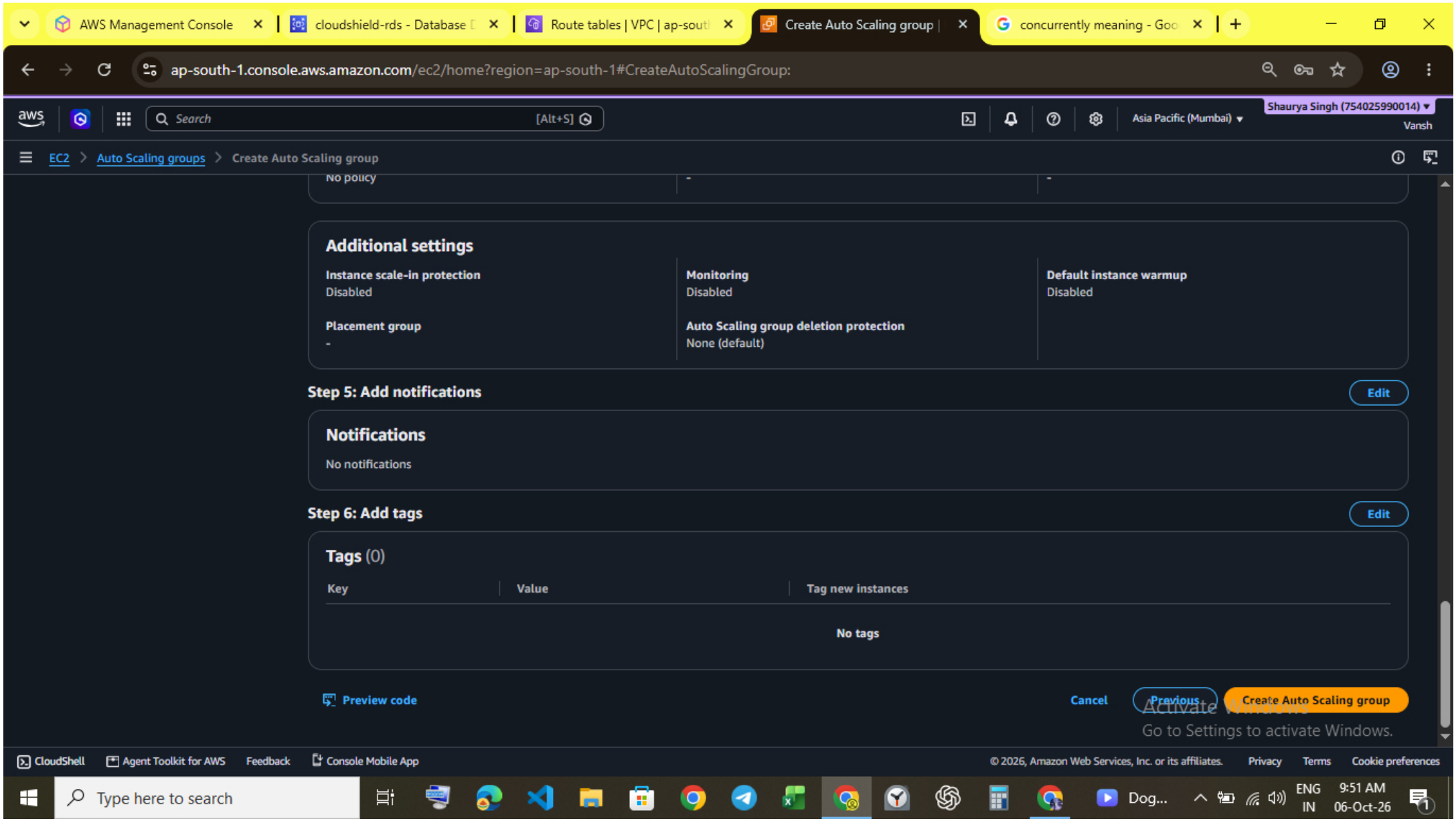

AWS Management Console
cloudshield-rds - Database
Route tables | VPC | ap-sout
Create application load bala
concurrently meaning - Goo
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateALBWizard:
Search
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EC2
Load balancers
Create Application Load Balancer
Review
Review the load balancer configurations and make changes if needed. After you finish reviewing the configurations, choose Create load balancer.
Summary
Review and confirm your configurations. Estimate cost
Basic configuration Edit
Name: cloudshield-alb
Scheme: Internet-facing
IP address type: IPv4
Network mapping Edit
VPC: vpc-00d60f9c997388d02
Public IPv4 IPAM pool: -
Availability Zones and subnets:
ap-south-1a
subnet-0494a741218e261a5
public-1
ap-south-1b
subnet-05dcae9e4b4b9b25f
public-2
Security groups Edit
ALB-SG
sg-01cbe28452ffa57d9
Listeners and routing Edit
HTTP:80 | Forward to 1 target group
Service integrations Edit
Amazon CloudFront + AWS Web Application Firewall (WAF): -
AWS WAF: -
AWS Global Accelerator: -
Tags Edit
-
Attributes
Certain default attributes will be applied to your load balancer. You can view and edit them after creating the load balancer.
Activate Windows
Go to Settings to activate Windows.
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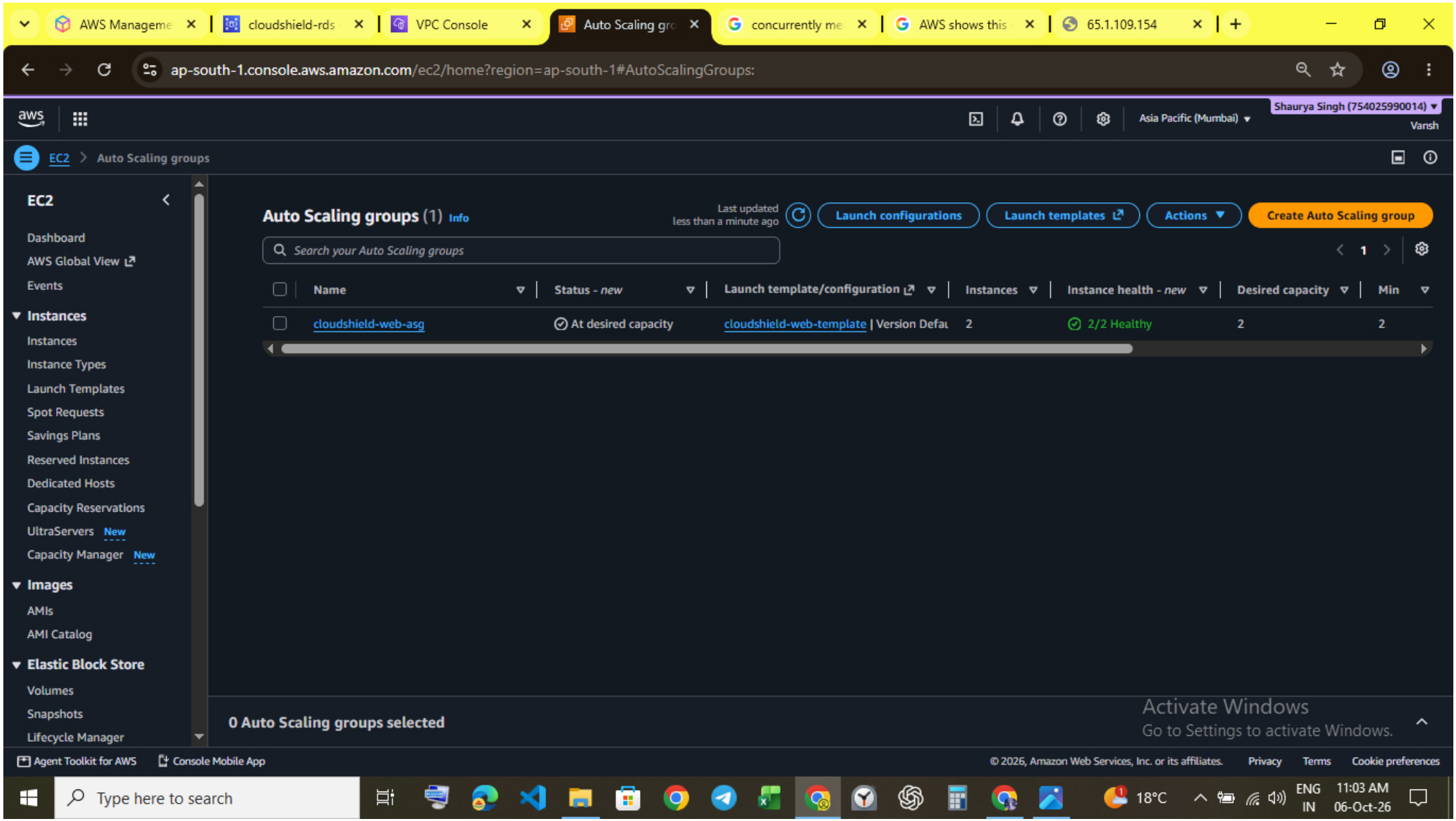

AWS Management Console
cloudshield-rds - Database
Route tables | VPC | ap-sout
Create Auto Scaling group
concurrently meaning - Goo
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateAutoScalingGroup:
Search
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EC2 > Auto Scaling groups > Create Auto Scaling group
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Step 1: Choose launch template
Edit
Group details
Auto Scaling group name
cloudshield-web-asg
Launch template
Launch template
cloudshield-web-template
lt-0fdf55cef0505b3f7
Version
Latest
Description
this template tells what type of instances to launch in order for the frontend to work properly.
Step 2: Choose instance launch options
Edit
Network
VPC
vpc-00d60f9c997388d02
Availability Zones and subnets
Availability Zone
Subnet
Subnet CIDR range
aps1-az1 (ap-south-1a)
subnet-0494a741218e261a5
10.0.1.0/24
aps1-az3 (ap-south-1b)
subnet-05dcae9e4b4b9b25f
10.0.4.0/24
Availability Zone distribution
Balanced best effort
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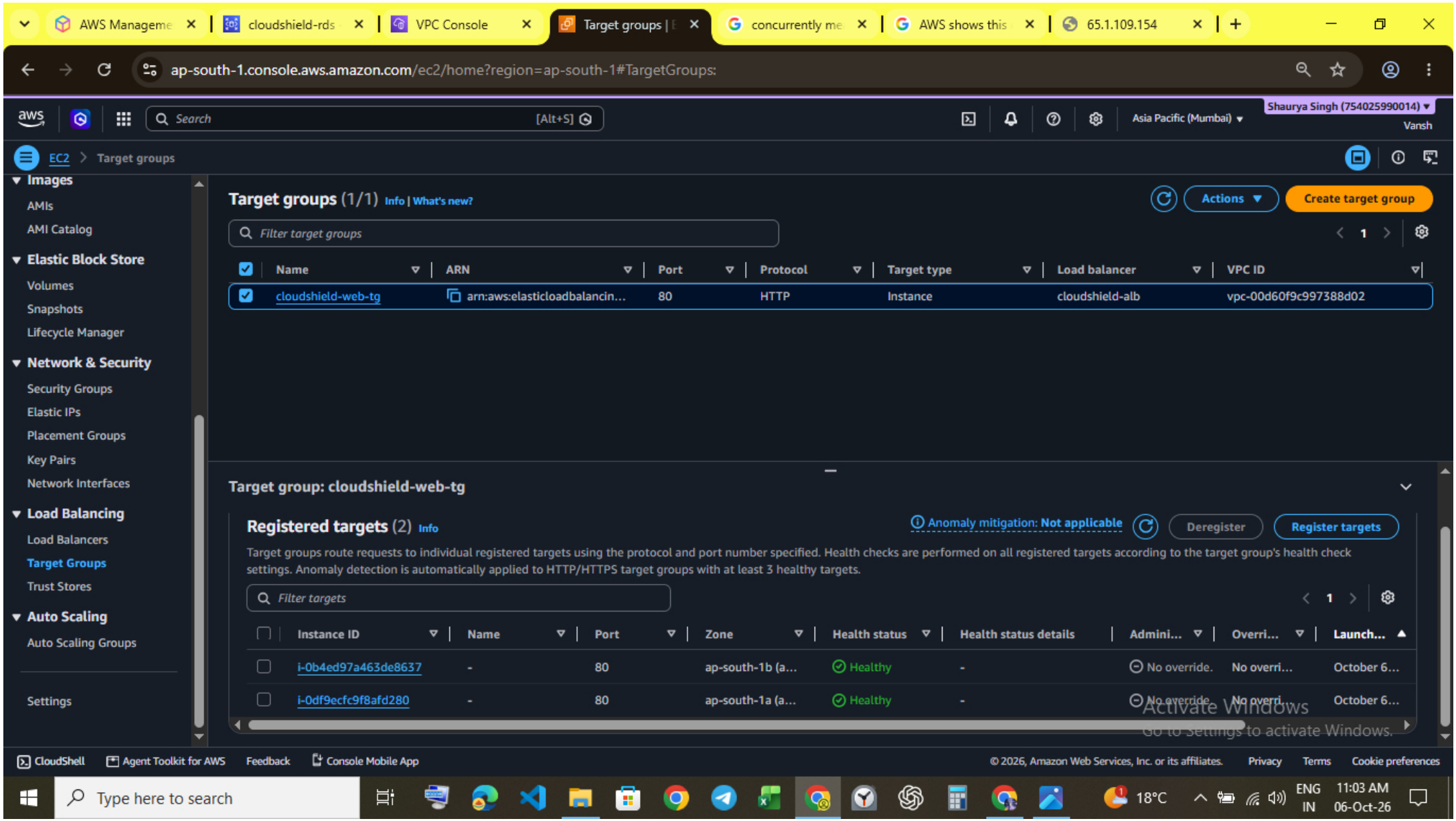

AWS Management Console
cloudshield-rds - Database
Route tables | VPC | ap-sout
Create Auto Scaling group
concurrently meaning - Goo
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateAutoScalingGroup:
Search
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EC2
Auto Scaling groups
Create Auto Scaling group
Capacity Reservation preference
Preference
Default
Capacity Reservation IDs
-
Resource Groups
-
Instance type requirements
This Auto Scaling group will adhere to the launch template.
Step 3: Integrate with other services
Edit
Load balancing
Load balancer 1
Name
cloudshield-alb
Type
Application/HTTP
Target group
cloudshield-web-tg
VPC Lattice integration options
VPC Lattice target groups
-
Application Recovery Controller (ARC) zonal shift
ARC zonal shift
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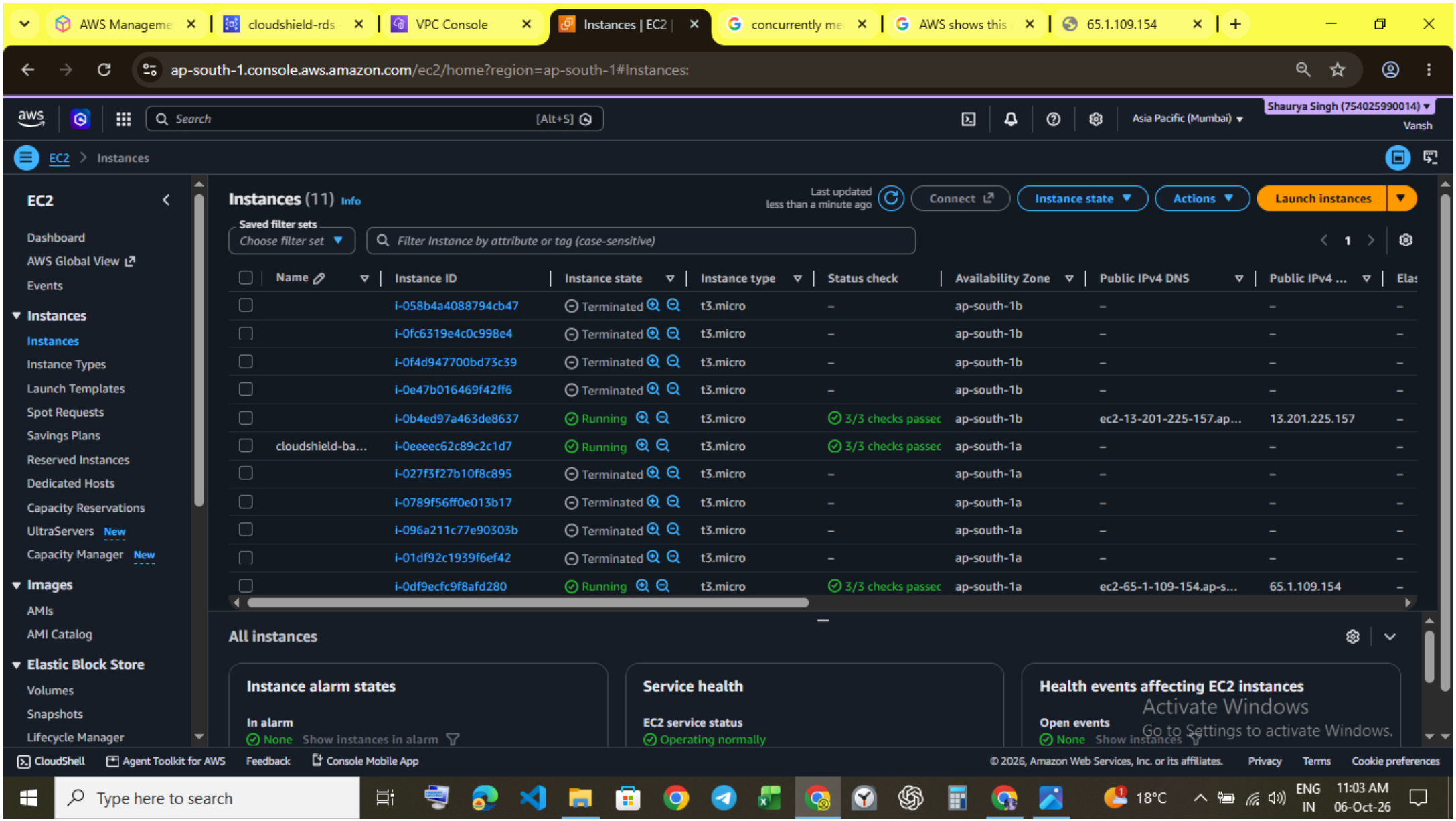

AWS Management Console
cloudshield-rds - Database
Route tables | VPC | ap-south
Create Auto Scaling group |
concurrently meaning - Goo
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateAutoScalingGroup:
Search
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EC2
Auto Scaling groups
Create Auto Scaling group
Application Recovery Controller (ARC) zonal shift
ARC zonal shift
Disabled
Health checks
Health check type
EC2, ELB
Health check grace period
300 seconds
Step 4: Configure group size and scaling policies
Edit
Group size
Desired capacity
2
Desired capacity type
Units (number of instances)
Scaling
Minimum desired capacity
2
Maximum desired capacity
3
Target tracking policy
Policy type
Target tracking scaling
Scaling policy name
Target Tracking Policy
Execute policy when
As required to maintain Average CPU utilization at 60
Take the action
Add or remove capacity units as required
Instances need
300 seconds to warm up before including in metric
Scale in
Enabled
Activate Windows
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AWS Management Console
cloudshield-rds - Database
Route tables | VPC | ap-south
Create Auto Scaling group
concurrently meaning - Goo
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateAutoScalingGroup:
Search
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EC2
Auto Scaling groups
Create Auto Scaling group
Target tracking policy
Policy type
Target tracking scaling
Scaling policy name
Target Tracking Policy
Execute policy when
As required to maintain Average CPU utilization at 60
Take the action
Add or remove capacity units as required
Instances need
300 seconds to warm up before including in metric
Scale in
Enabled
Instance maintenance policy
Replacement behavior
No policy
Min healthy percentage
-
Max healthy percentage
-
Additional settings
Instance scale-in protection
Disabled
Monitoring
Disabled
Default instance warmup
Disabled
Placement group
-
Auto Scaling group deletion protection
None (default)
Step 5: Add notifications
Edit
Notifications
No notifications
Activate Windows
Go to Settings to activate Windows.
Step 6: Add tags
Edit
CloudShell
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cloudshield-rds - Database
Route tables | VPC | ap-sout
Create Auto Scaling group |
concurrently meaning - Goo
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateAutoScalingGroup:
Search
[Alt+S]
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EC2
Auto Scaling groups
Create Auto Scaling group
No policy
Additional settings
Instance scale-in protection
Disabled
Monitoring
Disabled
Default instance warmup
Disabled
Placement group
-
Auto Scaling group deletion protection
None (default)
Step 5: Add notifications
Edit
Notifications
No notifications
Step 6: Add tags
Edit
Tags (0)
Key
Value
Tag new instances
No tags
Preview code
Cancel
Previous
Create Auto Scaling group
Activate Windows
Go to Settings to activate Windows.
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AWS Manageme
cloudshield-rds
VPC Console
Auto Scaling gro
concurrently me
AWS shows this
65.1.109.154
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#AutoScalingGroups:
Asia Pacific (Mumbai)
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EC2 > Auto Scaling groups
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Auto Scaling groups (1) Info
Last updated less than a minute ago
Launch configurations
Launch templates
Actions
Create Auto Scaling group
Search your Auto Scaling groups
Name
Status - new
Launch template/configuration
Instances
Instance health - new
Desired capacity
Min
cloudshield-web-asg
At desired capacity
cloudshield-web-template | Version Defa
2
2/2 Healthy
2
2
0 Auto Scaling groups selected
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Go to Settings to activate Windows.
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cloudshield-rds
VPC Console
Target groups | E
concurrently me
AWS shows this
65.1.109.154
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#TargetGroups:
Search
[Alt+S]
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Vansh
EC2 > Target groups
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Target groups (1/1) Info | What's new?
Actions
Create target group
Filter target groups
Name
ARN
Port
Protocol
Target type
Load balancer
VPC ID
cloudshield-web-tg
arn:aws:elasticloadbalancin...
80
HTTP
Instance
cloudshield-alb
vpc-00d60f9c997388d02
Target group: cloudshield-web-tg
Registered targets (2) Info
Anomaly mitigation: Not applicable
Deregister
Register targets
Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.
Filter targets
Instance ID
Name
Port
Zone
Health status
Health status details
Admini...
Overri...
Launch...
i-0b4ed97a463de8637
80
ap-south-1b (a...
Healthy
No override.
No overri...
October 6...
i-0df9ecfc9f8afd280
80
ap-south-1a (a...
Healthy
No override.
No overri...
October 6...
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cloudshield-rds
VPC Console
Instances | EC2
concurrently me
AWS shows this
65.1.109.154
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:
Search [Alt+S]
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Vansh
EC2 > Instances
EC2
Dashboard
AWS Global View
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Instances
Instance Types
Launch Templates
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Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Instances (11) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Public IPv4 ...
i-058b4a4088794cb47 Terminated t3.micro – ap-south-1b – –
i-0fc6319e4c0c998e4 Terminated t3.micro – ap-south-1b – –
i-0f4d947700bd73c39 Terminated t3.micro – ap-south-1b – –
i-0e47b016469f42ff6 Terminated t3.micro – ap-south-1b – –
i-0b4ed97a463de8637 Running t3.micro 3/3 checks passec ap-south-1b ec2-13-201-225-157.ap... 13.201.225.157
cloudshield-ba... i-0eeeec62c89c2c1d7 Running t3.micro 3/3 checks passec ap-south-1a – –
i-027f3f27b10f8c895 Terminated t3.micro – ap-south-1a – –
i-0789f56ff0e013b17 Terminated t3.micro – ap-south-1a – –
i-096a211c77e90303b Terminated t3.micro – ap-south-1a – –
i-01df92c1939f6ef42 Terminated t3.micro – ap-south-1a – –
i-0df9ecfc9f8afd280 Running t3.micro 3/3 checks passec ap-south-1a ec2-65-1-109-154.ap-s... 65.1.109.154
All instances
Instance alarm states
In alarm
None Show instances in alarm
Service health
EC2 service status
Operating normally
Health events affecting EC2 instances
Open events
None Show instances
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Go to Settings to activate Windows.
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